

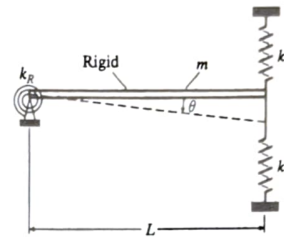
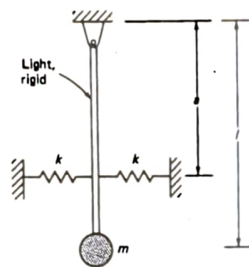
EGM 5421: Structural Dynamics - EXAM I

Instructions:

- (1) The test is open book and open notes. You may work on these problems at your home or any convenient place.
- (2) You may not discuss this test with anyone (from your class or outside). If you are unclear about a problem, you may contact me via email. **I will answer questions received in email by 5pm Friday, February 28, 2025.**
- (3) The test should be turned in by 11:50 p.m. on Sunday, March 2, 2025.

Name: MOSTAFA SAMEER

1. For the systems shown below, write the equations of motion. Assume small deformations.



2. A generator is placed on a base supported by spring-dampers. Spring stiffness is 80 kips/inch and the damping ratio is estimated to be 0.2. The weight of the system is 2240 lb. At an operating speed of 2000 rpm, the maximum amplitude of the vertical motion is 0.0025 inch. Find the force transmitted to the foundation, assuming that the motion is simple harmonic.
3. A cellular telecommunication tower is modeled as an SDOF. To identify the properties, a lateral force of 200 lb is applied at the top of the tower resulting in a horizontal displacement of 1 inch. A cable attached to the top of the tower is suddenly severed resulting in a free vibration. At the end of 2 complete vibration cycles, the time is 1.25 sec and the amplitude is 0.64 inch. Find effective stiffness, effective mass, undamped natural frequency and damping ratio for the SDOF system.
4. A frictionally damped SDOF system is undergoing free vibration after an initial displacement of 6-inch. The weight of the system is 1 kip, stiffness is 2 kips/inch and the friction coefficient is 0.15. Determine the displacement amplitude after 4 cycles and the number of cycles before the system comes to rest. Initial velocity may be assumed to be zero.
5. A simply supported W24x55 steel beam of span 40 ft supports a weight of 20 kips at its mid span. Upon free vibration, the system exhibits an amplitude decay of 0.75% per cycle. Find the equivalent viscous damping coefficient for the beam. Young's modulus of beam is 29000 ksi.

(a). A mass spring damper system with components on the right.

Equation of Motion:

$$m\ddot{x} + c\dot{x} + kx = 0.$$

where, m = mass;

c = damping coefficient;

k = stiffness;

x = displacement.

(b). A rigid bar pivoted at the one end with rotational damping and stiffness.

Equation of Motion:

$$I\ddot{\theta} + c\dot{\theta} + k\theta = 0$$

where, I = moment of Inertia;

c = rotational damping coefficient;

k = rotational stiffness;

θ = angular displacement.

Given,

A generator supported by spring dampers with following

Parameters

Spring Stiffness, $k = 80 \text{ kips/inch}$ (converted to $80,000 \text{ lb/inch}$).

Damping Ratio, $\xi = 0.2$

Weight of the System, $W = 2,240 \text{ lb}$. (converted to mass, $m = \frac{2,240}{386.4} = 5.8 \text{ lb-s}^2/\text{inch}$)

Operating Speed, $N = 2,000 \text{ rpm}$. (converted to $\omega = \frac{2000 \times 2\pi}{60} \approx 209.44 \text{ rad/s}$)

Maximum amplitude of vertical motion,
 $X = 0.0025 \text{ inch}$.

1. Calculate the natural frequency, ω_n

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{80,000}{5.8}} \approx 117.45 \text{ rad/s.}$$

2. Calculate the frequency ratio, r ;

$$r = \frac{\omega}{\omega_n} = \frac{209.44}{117.45} \approx 1.783$$

3. Calculate the term $2\xi r$;

$$2\xi r = 2 \times 0.2 \times 1.783 \approx 0.713$$

4. Calculate the force transmitted to the foundation

$$\begin{aligned} F_{\text{trans}} &= X \times k \times \sqrt{(1 + (2\xi r)^2)} = 0.0025 \times 80,000 \times \sqrt{1 + 0.713^2} \\ &= 200 \times 1.228 \\ &= 245.7 \text{ lb} \approx 246 \text{ lb.} \end{aligned}$$

Answer to the Question no: 03.

(03)

1. Effective Stiffness (k):

A lateral force of 200 lb results in a displacement of 1 inch. Stiffness $k = \frac{F}{x} = \frac{200 \text{ lb}}{1 \text{ inch}} = 200 \text{ lb/inch}$.

2. Damping Ratio (ξ):

After 2 complete vibration cycles (1.25 seconds), the amplitude is 0.64 inches

Logarithmic decrement, $\delta = \frac{1}{2} \ln$.

$$\left(\frac{1}{0.64}\right) = \frac{1}{2} \ln(1.5625) \approx 0.223.$$

$$\text{Damping Ratio } (\xi) = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}} \approx \frac{0.223}{\sqrt{39.4784 + 0.0498}} \\ \approx 0.0355 \approx \boxed{3.55\%}$$

3. Undamped Natural frequency (ω_n):

$$\text{Damped Period, } T_d = \frac{1.25 \text{ Sec}}{2} = 0.625 \text{ Sec.}$$

$$\text{Damped Natural Frequency, } \omega_d = \frac{2\pi}{T_d} \approx 10.053 \text{ rad/sec.}$$

$$\text{Undamped Natural Frequency, } \omega_n = \frac{\omega_d}{\sqrt{1 - \xi^2}} \approx \frac{10.053}{0.99937} \approx 10.06 \text{ rad/s}$$

4. Effective Mass

Converting Stiffness to lb/ft:

$$k = 200 \text{ lb/in} \times 12 = 2,400 \text{ lb/ft}.$$

$$\therefore \text{Effective Mass, } m = \frac{k}{\omega_n^2} = \frac{2,400}{(10.06 \text{ rad/s})^2} \approx 23.71 \text{ slugs}$$

1. Friction Force (F):

calculating friction force using friction coefficient and weight

$$F = \mu * W = 0.15 * 1 \text{ kip} = 0.15 \text{ kips.}$$

2. Amplitude Reduction Per Cycle:

The amplitude decreases linearly by:

$$\Delta X = \frac{4F}{k} = \frac{4 * 0.15 \text{ kips}}{2 \text{ kips/inch}} = 0.3 \text{ inches/cycle}$$

3. Displacement after 4 cycles:

Starting from 6 inches; Subtracting the total reduction after 4 cycles:

$$X_4 = 6 - (4 * 0.3) = 6 - 1.2 = \boxed{4.8 \text{ inches.}}$$

4. Cycles until rest:

Motion stops when the amplitude $X \leq \frac{F}{k}$:

$$\frac{F}{k} = \frac{0.15}{2} = 0.075 \text{ inches.}$$

for n:

$$6 - 0.3n \leq 0.075 \Rightarrow n \geq \frac{5.925}{0.3} = 19.75 \Rightarrow \boxed{20}$$

Answer:

(i) Displacement_{amplitude} after 4 cycles = 4.8 inches

(ii) Number of Cycles until rest = 20

1. Calculating the logarithmic decrement (δ):

The amplitude decay per cycle is 0.75%. So, the ratio of amplitudes between two consecutive cycles is 0.9925.

The logarithmic decrement is

$$\delta = \ln \left(\frac{1}{0.9925} \right) \approx 0.007528$$

2. Determining the damping ratio (ξ):

For small damping ratios; $\delta \approx 2\pi\xi$.

$$\text{for } \xi \rightarrow \xi \approx \frac{\delta}{2\pi} \approx \frac{0.007528}{6.2832} \approx 0.001198$$

3. Calculating the Stiffness (k) of the beam

The stiffness for a simply supported beam with

a central load is

$$k = \frac{48EI}{L^3}$$

$$k = \frac{48 \times 29,000,000 \text{ psi} \times 1,350 \text{ in}^4}{(480 \text{ in})^3}$$

$$= 17,000 \text{ lb/in}$$

$$= 17 \text{ kips/in}$$

Given,

$$E = 29,000 \text{ ksi}$$

(Young's Modulus)

$$I = 1,350 \text{ in}^4$$

$$L = 480 \text{ in}$$

4. Calculating the natural frequency (ω_n):

67

The mass 'm' is derived from the 20 kip weight:

$$m = \frac{W}{g} = \frac{20,000 \text{ lb}}{386.4 \text{ in/s}^2} \approx 51.75 \text{ lb}\cdot\text{s}^2/\text{in}.$$

The natural frequency:

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{17,000 \text{ lb/in}}{51.75 \text{ lb}\cdot\text{s}^2/\text{in}}} \approx 18.12 \text{ rad/s}$$

5. Calculating the equivalent viscous damping

coefficient (c):

Using the damping ratio (ξ); mass m; and

natural frequency ω_n :

$$c = 2 \cdot \xi \cdot m \cdot \omega_n.$$

$$= 2 \cdot 0.001198 \cdot 51.75 \cdot 18.12$$

$$\approx \boxed{2.25 \text{ lb}\cdot\text{s}/\text{in}}$$

The equivalent viscous damping coefficient for

the beam is $\boxed{2.25 \text{ lb}\cdot\text{s}/\text{in}}$